

Item Definition – Autonomous Emergency Braking (AEB)

1. Item Overview

1.1 Item Name

Autonomous Emergency Braking (AEB) System

1.2 System Purpose

The purpose of the AEB system is to **autonomously detect situations in which a forward collision is likely or imminent** and to **command vehicle deceleration by automatic braking** in order to avoid or reduce the severity of a collision with objects located in the vehicle's predicted path.

The system supports the driver by:

- Monitoring the forward driving corridor
- Assessing the relative position and motion of relevant objects
- Determining whether emergency braking intervention is required
- Requesting or applying braking intervention without requiring driver initiation

1.3 High-Level Functionality

At a high level, the AEB system performs the following functions:

1. Environment Perception

- Detects and tracks objects ahead of the vehicle, such as vehicles, pedestrians, cyclists, and static obstacles, depending on item scope and sensor configuration.
- Estimates object distance, relative speed, and path relevance.

2. Situation Assessment

- Evaluates whether the host vehicle is on a collision course with a detected object.
- Calculates collision-related metrics such as time-to-collision, closing speed, and predicted overlap in path.

3. Intervention Decision

- Determines whether autonomous braking should be requested or initiated based on predefined activation logic and operating conditions.

4. Vehicle Deceleration Request

- Sends a braking request to the braking/vehicle dynamics system.
- May coordinate with powertrain torque reduction where applicable.

5. Driver Information / Status Indication

- Provides system availability, intervention status, or temporary unavailability information through the vehicle HMI, if included within the item scope.

2. Functional Boundaries of the AEB System

2.1 Included Functions (Inside the Item Boundary)

The AEB item includes the functions necessary to:

- Receive and process sensor and vehicle state inputs relevant to forward collision assessment
- Detect and classify forward-path objects within the intended operational scope
- Track object motion and estimate host-object collision relevance
- Determine whether autonomous emergency braking criteria are met
- Generate braking intervention requests
- Manage internal AEB operating states such as:
 - Off / unavailable
 - Standby / monitoring
 - Active intervention
 - Temporarily suppressed or degraded operation (if defined in item behavior)

Depending on the intended architecture, the item may also include:

- Sensor fusion logic
- Object path prediction
- Driver status indication related specifically to AEB availability or intervention

2.2 Excluded Functions (Outside the Item Boundary)

The following are outside the AEB item boundary unless explicitly allocated to the item in a particular vehicle program:

- Manual service brake actuation by the driver
- Detailed hydraulic, electromechanical, or electrohydraulic brake actuation implementation
- Steering intervention or evasive steering
- Adaptive Cruise Control (ACC)
- Forward Collision Warning (FCW) as a separate feature
- Lane keeping / lane centering / path planning beyond collision assessment
- Airbag deployment
- Occupant restraint control
- Post-crash handling functions
- General-purpose perception functions not used by AEB
- Map-based navigation and route planning
- V2X-based collision avoidance functions unless specifically part of the item definition

2.3 Triggering and Functional Scope

The AEB system is intended to operate in **forward-driving scenarios** in which:

- The vehicle is moving within a defined speed range
- Objects are within the detectable and assessable forward path
- Relevant environmental and system conditions allow the item to operate

The item scope may include one or more of the following scenario classes, depending on vehicle implementation:

- Vehicle-to-vehicle rear-end collision scenarios
- Vehicle-to-pedestrian forward crossing or longitudinal scenarios
- Vehicle-to-cyclist scenarios

- Stationary obstacle scenarios

If a scenario class is not explicitly supported, it is outside the functional scope of the item.

3. External Interfaces and Interacting Elements

3.1 External Interfaces

The AEB item interfaces with vehicle elements external to the item boundary. Typical external interfaces include:

a) Perception Sensors

The AEB system may receive input from one or more of the following:

- Front radar
- Front camera
- Front lidar (if applicable)
- Ultrasonic sensors (typically limited relevance for AEB at higher speed)
- Sensor fusion domain controller or perception ECU

Typical exchanged information:

- Object list
- Raw or processed detections
- Classification labels
- Relative distance and velocity
- Confidence or validity indicators

b) Vehicle Motion and State Signals

The AEB item may receive:

- Vehicle speed
- Longitudinal acceleration
- Yaw rate
- Steering angle

- Wheel speed signals
- Gear state / drive direction
- Accelerator pedal position
- Brake pedal status
- ABS/ESC status
- Powertrain torque status

These signals are used to determine current vehicle motion, driver demand, and intervention relevance.

c) Braking / Vehicle Dynamics System

The AEB item sends output to:

- Brake control module / ESC / integrated brake system
- Vehicle dynamics controller

Typical exchanged information:

- Requested deceleration
- Brake torque or pressure request
- Intervention enable/request flag
- Request timing / ramp profile (if part of system interface)
- Status feedback / acknowledgement from brake system

d) Powertrain System

Where applicable, the AEB item may interface with:

- Engine control unit
- Motor/inverter controller
- Hybrid control unit

Typical purpose:

- Request propulsion torque reduction or suppression during AEB intervention

e) Human-Machine Interface (HMI)

The AEB system may provide or receive:

- AEB availability indication
- AEB intervention indication
- Driver warning coordination (only if within item scope)
- System enable/disable configuration input, if user-configurable

f) Diagnostic and Communication Interfaces

The AEB item may interface with:

- Vehicle network (CAN, CAN FD, FlexRay, Automotive Ethernet, etc.)
- Diagnostic services for fault reporting, status, and service functions
- Central gateway or domain controller

3.2 Interacting Elements

The principal interacting elements are:

- Driver
- Host vehicle braking system
- Vehicle motion sensors and chassis systems
- Forward environment perception sensors
- Powertrain system
- HMI / instrument cluster
- Other ADAS or motion control functions that may share sensor or actuator resources

The driver remains an interacting external element because:

- The driver may steer, brake, or accelerate while the system is monitoring or intervening
- Driver inputs may influence the system's activation eligibility or continued intervention behavior, depending on implementation

4. Operational Environment and Operating Conditions

4.1 Operational Environment

The AEB item is intended for use on public or private roadways within the vehicle's intended use domain, including:

- Urban roads
- Rural roads
- Highways / expressways
- Structured traffic environments with forward travel direction

The item is intended to function in normal traffic scenarios involving:

- Lead vehicles
- Stopped or slower-moving objects ahead
- Pedestrians or cyclists, if included in the item scope
- Lane-following or near-lane-following travel conditions

4.2 Operating Conditions

The AEB system operates subject to defined conditions including, but not limited to:

Vehicle Conditions

- Vehicle ignition / propulsion system active
- Vehicle traveling in a forward direction
- Vehicle speed within the item's specified operating range
- Required vehicle dynamic signals available and valid
- Brake system capable of accepting intervention requests

Environmental Conditions

- Daytime and nighttime conditions within sensor capability
- Road environments with varying curvature and grade within specified limits
- Ambient weather conditions such as clear, rain, fog, or low light, subject to sensor performance capability
- Road surface conditions such as dry or wet pavement, within the vehicle and brake system operating capability

Traffic Conditions

- Moving traffic
- Slower or stationary objects in path
- Cut-in or decelerating lead-object scenarios, if supported by the item definition
- Presence of vulnerable road users, if within item scope

System Conditions

- Relevant sensors operational and providing valid data
- Communication paths available
- Required upstream/downstream ECUs operational
- No active mode exclusion that prevents AEB operation

4.3 Mode of Operation

Typical AEB operating modes include:

- **Inactive/Off** – system not available due to configuration or vehicle state
- **Standby/Monitoring** – environment monitored and collision assessment active
- **Intervention** – braking request issued or active braking ongoing
- **Unavailable/Degraded** – function not available or capability reduced due to environmental, vehicle, or interface conditions

5. Assumptions and Limitations of the Item

5.1 Assumptions

The item definition assumes the following:

1. Sensor Availability

- Required perception sensors or upstream perception outputs are available, correctly installed, and functioning within specification.

2. Vehicle State Availability

- Vehicle motion and status signals used by the AEB algorithm are valid, timely, and communicated over the vehicle network.

3. Brake System Availability

- The brake system can receive and execute braking requests generated by the AEB item within the defined interface behavior.

4. Vehicle Maintenance and Calibration

- Sensors, actuators, tires, brakes, and chassis-related components are maintained according to vehicle requirements and remain within intended tolerances.

5. Road Use Context

- The host vehicle is being operated in a forward-driving road context consistent with the intended operational environment.

6. Object Detectability

- Objects relevant to AEB intervention are detectable within the physical and algorithmic capability of the available sensors and perception chain.

5.2 Limitations

The AEB item is subject to inherent functional limitations, including:

1. Limited Operational Design Scope

- The system operates only within defined speed ranges, object classes, path geometries, and scenario types.
- Scenarios outside the specified operational domain are not part of the item scope.

2. Sensor-Dependent Performance

- Performance depends on sensor field of view, range, occlusion, contamination, lighting, weather, and target characteristics.
- Certain objects or object trajectories may not be detected or correctly classified.

3. Path Relevance Constraints

- The item is primarily intended for forward-path collision scenarios and may have limited relevance for highly lateral, non-lane-aligned, or complex multi-object interactions.

4. **Vehicle Dynamics Constraints**

- Achievable deceleration depends on road-tire friction, vehicle loading, brake capability, and current vehicle dynamics.

5. **Driver Interaction Constraints**

- Driver inputs may affect item behavior depending on implementation, such as simultaneous braking, accelerator application, or steering maneuvers.

6. **Environmental Constraints**

- Performance may be limited in adverse weather, poor visibility, glare, low contrast scenes, unusual road geometry, or degraded lane/context cues where used by the algorithm.

7. **Interface Dependency**

- The item depends on proper operation of external ECUs, communication networks, and actuator interfaces that are outside the item boundary.

8. **Not a General Collision Avoidance System**

- The item is limited to the specific forward emergency braking function and should not be interpreted as providing full crash avoidance capability in all driving situations.

6. **Summary of Item Intent**

The AEB item is a **forward collision intervention function** that monitors the host vehicle's forward path, assesses collision likelihood with relevant objects, and autonomously requests braking when predefined intervention criteria are met. Its scope is bounded by:

- the supported scenario classes,
- available environmental perception,
- vehicle state inputs,
- actuator interface availability, and
- defined operational conditions.